

SANTAREM, Brazil

WMO: 822440

Lat: 2.422S

Lon: 54.793W

Elev: 60

StdP: 100.60

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	21.9	22.3	19.8	14.6	28.9	20.2	15.0	28.3	8.5	26.6	8.1	26.8	1.0	150	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	8.1	34.2	25.3	33.9	25.3	33.1	25.5	26.8	30.7	26.6	30.6	26.4	30.4	5.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.6	28.6	25.2	20.5	28.2	25.2	20.5	28.2	84.6	31.0	83.4	30.8	82.4	30.7	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.5	6.5	21.2	35.1	0.9	0.7	20.6	35.6	20.1	36.1	19.6	36.5	19.0	37.0
			20.3	27.8	0.9	1.1	19.7	28.7	19.1	29.3	18.6	30.0	18.0	30.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	27.6	27.1	27.1	27.1	27.1	27.1	27.2	28.0	28.9	29.1	28.9	28.5
	DBStd	1.37	1.53	1.30	1.19	1.05	0.93	0.83	0.82	0.85	0.93	1.17	1.20	1.45
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6504	544	479	531	514	531	514	533	558	566	593	567	573
	CDD18.3	3462	286	246	272	264	273	264	274	300	316	334	317	315
	CDH23.3	36856	2887	2295	2527	2467	2573	2549	2708	3403	3884	4141	3829	3592
	CDH26.7	14376	1033	710	757	735	794	892	1020	1462	1774	1936	1751	1513
Wind	WSAvg	2.9	3.0	2.9	2.8	2.6	2.5	2.4	2.4	2.9	3.4	3.5	3.4	3.3
Precipitation	PrecAvg	1960	215	263	369	339	258	136	104	44	30	38	63	101
	PrecMax	2589	392	437	642	536	520	244	207	157	94	105	159	266
	PrecMin	1151	32	76	178	211	115	41	27	13	3	8	9	2
	PrecStd	322	96	94	97	86	106	47	50	30	23	28	35	76
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	33.1	33.0	32.2	32.1	32.2	32.8	33.2	34.2	35.1	34.8	34.2
		MCWB	25.7	26.0	25.8	25.8	26.1	25.5	25.2	25.7	25.2	25.2	24.9	25.3
	2%	DB	33.0	32.0	31.8	31.2	31.2	31.9	32.0	33.0	33.7	34.1	34.0	33.9
		MCWB	25.5	25.6	25.6	25.8	26.0	25.6	25.3	25.7	25.3	25.3	25.2	25.3
	5%	DB	32.0	31.1	30.9	30.9	30.9	31.1	31.2	32.2	33.1	33.8	33.2	33.0
		MCWB	25.2	25.6	25.6	25.8	26.0	25.7	25.5	25.5	25.5	25.3	25.4	25.3
	10%	DB	31.0	30.1	30.1	30.1	30.1	30.8	31.0	32.0	32.8	33.0	32.9	32.1
		MCWB	25.4	25.4	25.5	25.6	25.7	25.7	25.6	25.4	25.5	25.4	25.4	25.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	26.9	26.8	27.0	27.1	26.7	26.7	26.8	26.8	26.7	26.7	26.7
		MCDB	30.2	30.7	30.1	30.3	30.3	30.5	30.6	31.1	31.4	31.9	31.5	30.9
	2%	WB	26.2	26.2	26.2	26.5	26.6	26.4	26.2	26.5	26.5	26.4	26.3	26.2
		MCDB	30.2	29.9	29.6	29.6	29.8	30.1	30.2	30.9	31.2	31.6	31.3	30.8
	5%	WB	26.0	26.0	26.0	26.1	26.2	26.1	26.1	26.2	26.2	26.2	26.1	26.0
		MCDB	30.1	29.7	29.4	29.2	29.4	29.9	30.1	30.8	31.0	31.2	31.1	30.7
	10%	WB	25.7	25.7	25.7	25.9	26.0	25.8	25.8	26.0	25.9	25.9	25.9	25.7
		MCDB	29.6	29.1	28.9	28.9	29.2	29.5	29.8	30.6	30.8	30.9	30.8	30.3
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.3	6.1	6.1	6.3	7.1	7.6	8.1	8.0	8.1	8.0	7.3
		MCDBR	7.5	7.1	6.9	6.8	7.0	7.7	8.1	8.4	8.3	8.8	8.5	7.9
		MCWBR	2.4	2.7	2.7	2.8	3.0	3.4	3.5	3.1	2.7	2.7	2.7	2.4
	5% WB	MCDBR	7.0	6.6	6.4	6.3	6.5	7.3	7.7	8.2	8.0	8.1	8.1	7.4
		MCWBR	2.7	2.7	2.8	2.8	3.0	3.4	3.5	3.3	2.9	2.8	2.9	2.7
Clear-Sky Solar Irradiance	taub	0.397	0.398	0.392	0.383	0.386	0.388	0.402	0.437	0.490	0.488	0.502	0.457	
	taud	2.533	2.540	2.562	2.610	2.586	2.575	2.512	2.373	2.211	2.229	2.182	2.335	
	Ebn at Noon	937	936	931	916	889	874	867	855	831	843	833	875	
	Edh at Noon	109	110	106	98	97	96	103	123	149	148	154	132	
All-Sky Solar Radiation	RadAvg	4.82	4.68	4.73	4.66	4.59	4.91	5.21	5.77	5.93	5.74	5.35	4.95	
	RadStd	0.25	0.21	0.26	0.24	0.28	0.21	0.24	0.14	0.18	0.24	0.23	0.39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page